

.NET deserialization attacks and their associated threats in the world of CMS - Daniel Susid

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Content

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Description

We will briefly cover the history of deserialization vulnerabilities and explain what they are, as well as describe how they can potentially expose websites using Umbraco Content Management System (CMS) to high-severity attacks. We will highlight typical scenarios in the context of a CMS and demonstrate some applicable techniques using concrete examples of .Net deserialization exploitation. We will then also look at some ways to discover and prevent .Net deserialization vulnerabilities. Finally, we will perform a hands-on exercise discovering, exploiting, and preventing a deserialization vulnerability in a CMS plugin.

Speaker: Daniel Susid

<https://codegarden.umbraco.com/>

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Transcript

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hello again that was the quickest break ever um I said before this is Daniel seussid and he's going to talk about .net there is the serialization attacks in the and their Associated threats in the world of content Management Systems so take away Daniel thanks I'll catch you later all right so my name is Daniel susid um I've been in the security industry for about 15 plus years something like that and currently I work as a security specialist at outpost24 and my focus right now is predominantly on offensive security so I do a lot of different stuff like penetration testing red teaming and similar stuff and application security is a big part of that also and this is the first time I'm in cold Garden so it's very interesting to see it's very nice and relaxed so this serialization attacks us oh okay great thank you all right so I don't know if anyone here is interested in security but this realization attacks are are very very common thing these days but it's not a new thing so it from a Java perspective it was already a concept in 2016. but James Forshaw introduced it in terms of net civilization attacks already in 2012 so it's been here for a long time actually and uh Alvaro Munoz actually introduced it or he did a lot of research in that area and he really put it on the center stage in 2017. so why talk about deserialization attacks there's been a lot of recent events with attacks in this area it's quite high level of complexity in terms of understanding these type of attacks and often they are quite misunderstood it's easy to think they're too complex they cannot happen to us or or we just don't understand that we're vulnerable and the actual impact of our attack is is very high I don't know if you recognize this anyone [Music] I put this here on a purpose because I think this really really made the headlines it was a really big thing right in the yoga World everyone knows lock for Jay everyone uses log4j and if you don't use log4j then you're probably not a coder and very soon after the first vulnerability came we had five six seven more Olympics is coming really really fast all concerning deserialization type of attacks but this is the yoga world right this is Java this is not .net and we're talking about .net civilization attacks so what

makes that interesting does anyone recognize this we had four zero days in Microsoft Exchange right so why is this interesting for one I was working in a sock at that time and it was crazy I I tell you I was I was afraid my manager was crying and things were happening all over the place but there is a specific purpose why I'm actually mentioning this so if you look at the exchange service blockchain here you can see on the center stage actually you have a net utilization vulnerability which used the attack used for the exchange server attack which was exploded by for example half in them a Chinese threat actor it was quite sophisticated and it was a very complex exploit chain consisting of several different exploits so you had a s you know uh service side request forgery type of attack going on and then our file right arbitrarily file right going on post Authentication and then they used actually.net deserialization to actually escalate to to system privileges so I would say this really puts dotnet on or net digitalization type of attacks on the map for sure so what about CMS right we're in the umbraco world we talk about cms's do you recognize these these are net Focus so I mean there are lots more in different languages in Java in PHP and so forth but if you just focus if you just look at the net side of things these are quite recent you have a deserialization type of of vulnerability in or in Orchestra which is a very popular CMS you have a pre-auth RC type of digitalization type of vulnerability in sitecore and and so on and so on so we can't really say that it's not a common thing right now right it's still happening so we have to be aware we have to keep it in mind I know everyone here is a developer so or maybe a manager post developer or whatever but still you you do know what serialization is but I'll still do a recap so we're basically talking about marshalling an object converting an object from you know from a structured object to a format that we maybe it's more human readable or it's easier to transfer over the network or store in the file system and so forth and then you have the reverse order right you have the deserialization part which is basically reconstructing the object and use it again as an instance so when you look at it I mean the flow is not too complicated but of course there's a lot of complexity complexity underneath and the big issue here from an attack perspective is the deserialization part right when you're actually reconstructing the object but digitalization doesn't make you know make your source vulnerable just by by using it you can implement digitalization in a secure way right and it's it's more a matter of actually how you use it and combination of factors how you configure the libraries that you use and if you consider other aspects if you are using untrusted data for for one so basically in terms of when we talk about insecure deserialization it means that we have certain conditions that allows a user to actually inject an object into the marshalling process or the desertization process and uh if you can do that the outcome is quite simply you know you can do a dialogue service type of attack you can do a remote code execution or you can just manipulate the logic in in various ways so the big thing is can the attacker control the object can the attacker control the type right and of course you in .net and so forth you need to have a decentralization gadget so-called right that's like a trampoline that can actually execute a method so if we look at it from a simple perspective this is a typical very common Library json.net and in this case we're actually setting the type name handling to to objects and you can find this in in a lot of places and if you look at the open source code basis you see this a lot it's very common and because it's easy the code you know you you when you want to be serious an object it automatically

determines the type so it makes things very easy but there is a pitfall here what happens when you can automatically determine the type from the actual data that you are consuming and deserializing and if an attacker can actually control that data then you have our deserialization issue then you have insecure deserialization this is a very normal typically extracted Json object for example if you would deserialize the object it would look pretty similar to this but there are some things here which are a little bit different and if you look at this object this is actually a malicious object in terms of if an attacker would send this to their previous method which has type name handling Etc objects which means it actually determines the type it would actually instantiate not the type that maybe was set in the code from the beginning or expected in the code but to the type actually specified in the actual Json and there's another aspect here if you look at it you have something called the object data provider right so that's a wrapper class and that wrapper class we call that or usually call it a gadget right and that Gadget we can use to actually execute another type of of of instance method for example in this case we are executing the the start method of the system diagnostic process class you couldn't do this you couldn't just instantiate the system usually process and execute the start method by itself you need to have a gadget right you need to have a magic so to say class or magic function and the object that I provided class is quite special in that way because it actually as soon as it actually gets the object instance the dope you can say set for example like in this case the system diagnostic process it will after instantiating that actual class called the the method name which is provided here the start and provide the method parameters so in this case we can actually execute code while performing deserialization so the code I just showed you a very simple example this one here where you actually are setting type name handling to objects when performing deserialization was actually found very recently just about a year ago 2021 in one of our most popular cms's Orchestra C1 and you can see a little bit of a a screenshot here of the actual code and how it's set so it's set to type name handling equals four right so that is the same as type name handling dot objects right so a very simple thing like this made or created a vulnerability in the CMS which actually allowed executing remote code on the server post off though you needed to be authenticated but still you can actually execute code on the server just by sending a request that will be deserialized an interesting aspect actually in terms of this is also when you look at this code base they're actually using something which is called the composite serialization binder here and one of the use your recommendations to address the issue with digitalization and and types and so on and restricting the types that can be deserialized is to use a civilization binder however in this case which is quite interesting they did actually implement the civilization binder but they only restricted the actual assembly names or type names or namespaces that conform to their namespace anything else outside of that was okay and was completely unrestricted so you could pretty much just realize anything that you wanted so if we look at the hypothetical attack scenario in the context of a CMS I'm sure today everyone is pretty accustomed to this but if we just look at it from as a from a amusement point of view so as an attacker right so you have a vulnerability we mentioned like side core before these are pre-off vulnerabilities you also have the kentico CMS vulnerability from 2019 as an example dot-net digitalization pre-off it's extremely easy for an attacker to cause huge amounts of damage and there's a lot of

information there's a lot of threat intelligence about threat actors actually using the civilization type of attacks to gain the initial foothold in a corporate environment it's very very common so if you look at this scenario the attacker would basically just attack it's pre-auth he doesn't need to be authenticated he can just send a request his in is on the server is executing code with the same privileges as the server and he can then escalate to a system perform lateral movement and do whatever he really wants within the company right so it has a lot of implications it's there should be a lot of awareness about this so if we look at another hypothetical attack scenario for example if you look at the pre sorry the post off type of vulnerability that we have for example in the orchestra CMS so we have one from two from 2021 and we have another one from 2019 right so the two the one from 2021 is the one actually with the type name handling configuration set a little bit in a strange way so if you look at it from from this perspective you can see that it's not that difficult to abuse this either it's there is a little bit of a higher threshold here for an attacker but still you can really abuse this for example as an attacker you could obtain leaked credentials so you can actually access the server it doesn't have to be high privileged credentials it can be low privileged credentials so you can actually access the CMS and then you can actually perform post off type of desertization attacks and get access to the server but you can also perform this kind of attack which is post off through a csrf type of attack cross-site request forgery attack and you can combine that with social engineering so when you actually combine this kind of post-auth vulnerabilities with other type of of issues you get a very complex sometimes complex but very efficient and very effective exploiting and in this case for example it wouldn't take much for the attacker to actually be able to abuse this to get a foothold within the the infrastructure just like in the in the pre-auth type of attack and the same type of procedure you get access to the server you're inside the parameter you can do privilege escalation to system and then start doing various types of lateral movement through active directory or whatever so how do we prevent this kind of things well one thing is to avoid the serializing and trusted data I mean of course we're going to be using digitalization right it's it's really a really nice feature and it it's really helpful for many things but we should try to avoid this realizing untrusted data as much as possible but if we really need to disable untrusted data we should really try to evaluate the security of the libraries that we use and we should ensure that the libraries that we use that they have strict type control it's also important to try to ensure that the library specific options that we use for example like the type name handling that we set we looked at before that it's actually set according to best practice and also try to consider various implications of various configuration settings it can get complicated sometimes that's why white box reviews code reviews are really important it's not always easy to find these things through just static analysis it's also important to do Dynamic analysis and various code reviews overall so a combination of these three aspects is really good and one thing which is really really important never use user control data to actually Define the type which is expected by the actual deserializer this is also another aspect that we see which is actually quite common to be honest even when we talk about diff different kind of formatters not only json formatters like json.net it can be XML serializer whatever binary formatter and so forth another aspect is to use class whitelists use serialization binders but when you do be very careful and review the implementations and as I said go back to the

beginning if you can avoid visualization of untrusted data that's the key we're going to be looking we're going to have a workshop on Friday or we're going to be going through these uh RCS here for example the orchestra C1 CMS RC and another one also this one is the from to from 2021 we're also going to go through the one from 2019. and we're going to actually be stepping through going through the whole code base we're going to be performing and and creating our own test payloads and Performing various types of attacks to try to see how to abuse this and also a little bit in terms of how we actually would approach it from a perspective when we only have a closed source project in terms of actually reverse engineering the actual uh software at hand there is a lot of other information you can get oasp has great cheat sheets you can read a lot about digitalization issues and how to avoid them and how to actually stay safe as I mentioned Alvaro Munoz before he is really really one of the leading people in this area and he has a lot of nice information that he has written out and research published and you can read a little bit about what James forsha already mentioned in 2012. and yeah that's it thank you [Applause]